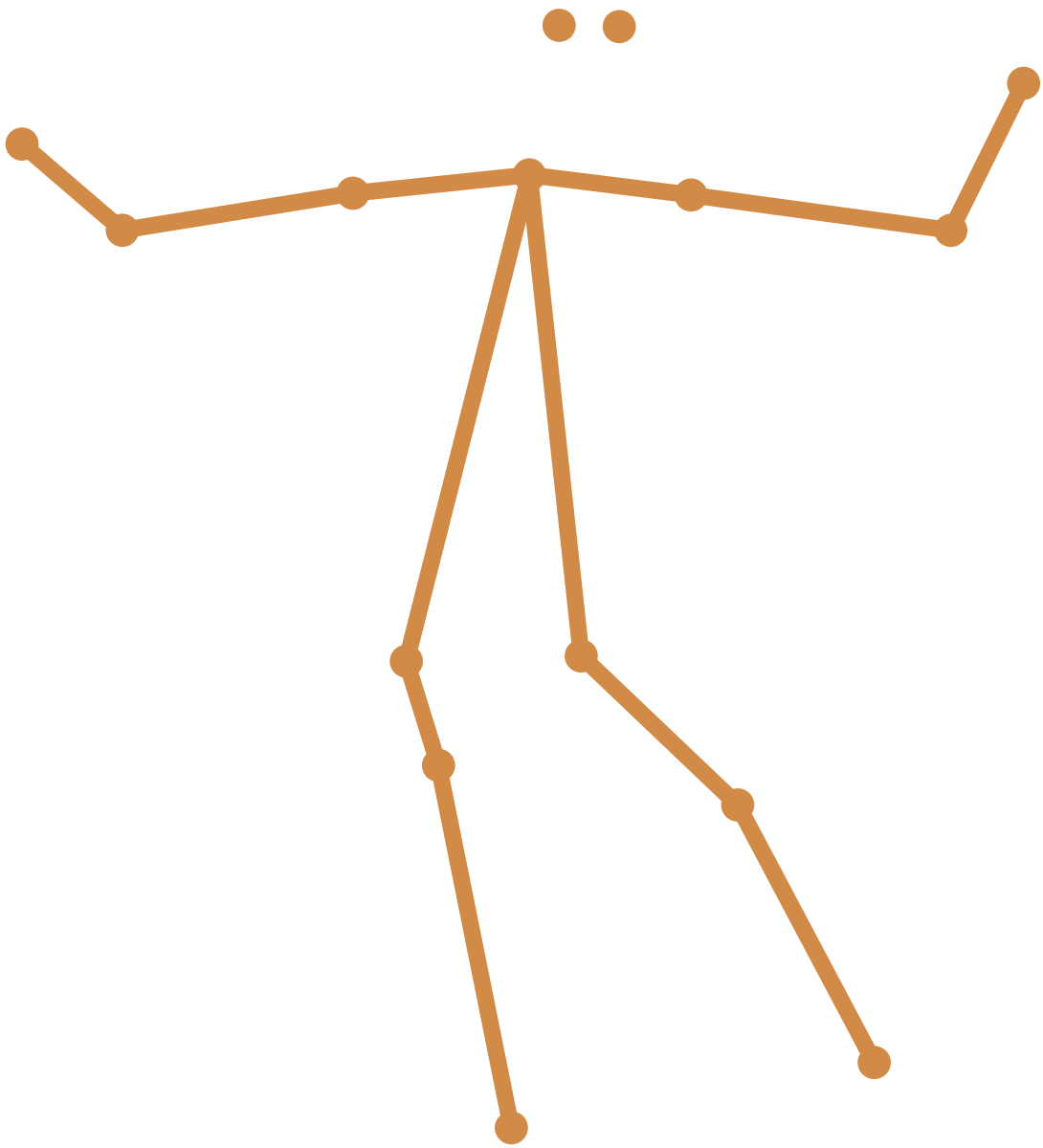Refined canonical pose